Q4 — Inventory Database

Low Stock Products With Supplier Information

📁 Database Description

The inventory database manages products, suppliers, categories, and stock information. Products are associated with suppliers, and the inventory system tracks the quantity currently available for each product.

📑 Schema Details

Tables:

- Products
- Suppliers
- Inventory

📌 Problem Statement

The inventory team wants to identify products that are currently running **low on stock**. For each low-stock product, display the product name, supplier name, current stock quantity, and reorder level.
Consider a product to be low on stock when:
stock_quantity < reorder_level

📌 Query to be Implemented

Write a query to return:

- product_id
- product_name
- supplier_name
- stock_quantity
- reorder_level

for products whose current stock is below their reorder level.

📌 Expected Output

product_id	product_name	supplier_name	stock_quantity	reorder_level
101	Wireless Mouse	TechSupply	8	20
105	USB Keyboard	Computer World	12	25

SQL concepts tested: SELECT, JOIN, WHERE, column selection.

Q4 — SmartMart Database

Completed Orders With Customer Information

📁 Database Description

The smartmart database manages retail operations including customers, products, orders, and order transactions. Customers place orders, and each order is maintained with a status indicating its current state.

📑 Schema Details

Tables:

- Customers
- Orders

📌 Problem Statement

The SmartMart sales team wants to view all orders that have been **successfully completed**. For each completed order, display the customer name, order date, and order status.
Only orders having:
status = 'Completed'
should be included.

📌 Query to be Implemented

Write a query to return:

- order_id
- order_date
- customer_name
- status

for all completed orders.

📌 Expected Output

order_id	order_date	customer_name	status
1001	2025-08-12	Rahul Sharma	Completed
1005	2025-08-15	Priya Nair	Completed
1010	2025-08-18	Arjun Kumar	Completed

SQL concepts tested: JOIN, WHERE, filtering using a string condition.

Q4 — AdventureWorks Database

Products With Their Category Information

📁 Database Description

The AdventureWorks database contains information about products, product categories, product subcategories, and sales transactions. Products are organized into categories and subcategories.

📑 Schema Details

Tables:

- Production.Product
- Production.ProductSubcategory
- Production.ProductCategory

📌 Problem Statement

The product management team wants to identify products that belong to the **Bikes** category. For each such product, display the product name, product number, subcategory name, and category name.
Only products belonging to the **Bikes** category should be displayed.

📌 Query to be Implemented

Write a query to return:

- ProductID

- Name
- ProductNumber
- SubcategoryName
- CategoryName

for products belonging to the Bikes category.

📌 **Expected Output**

ProductID	Name	ProductNumber	SubcategoryName	CategoryName
680	HL Road Frame - Black, 58	FR-R92B-58	Road Frames	Bikes
706	HL Road Frame - Red, 58	FR-R92R-58	Road Frames	Bikes
712	HL Road Tire	TI-R092	Road Bikes	Bikes

SQL concepts tested: multiple JOINs, filtering, table aliases, selecting columns from different tables.

Q4 — Student Database

Students Enrolled in a Specific Course

📁 Database Description

The student database manages student information, courses, and student enrollments. Students can enroll in multiple courses, and each enrollment connects a student with a particular course.

📑 Schema Details

Tables:

- Students
- Enrollments
- Courses

📌 Problem Statement

The academic department wants to identify all students who are currently enrolled in the **Database Management Systems** course.
For each enrolled student, display the student ID, student name, course name, and enrollment date.

📌 Query to be Implemented

Write a query to return:

- student_id
- student_name
- course_name
- enrollment_date

for students enrolled in the **Database Management Systems** course.

📌 Expected Output

student_id	student_name	course_name	enrollment_date
101	Ananya Rao	Database Management Systems	2025-07-10

104	Rahul Verma	Database Management Systems	2025-07-11
109	Sneha Patel	Database Management Systems	2025-07-13

SQL concepts tested: multiple JOINs, WHERE, filtering based on course name.

Pattern Summary

These four questions deliberately follow the **same difficulty and structure as your PDF**, rather than jumping immediately to complex SQL:

Database	Scenario	Main SQL Skill
Inventory	Low-stock products + supplier	JOIN + comparison WHERE
SmartMart	Completed orders + customer	JOIN + string filtering
AdventureWorks	Products + subcategory + category	Multiple JOINs + filtering
Student	Students + enrollment + course	Multiple JOINs + filtering